

		BF600 Data Form			
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1. Bluetooth

The BF600 is able to transfer stored measurement data via Bluetooth Low energy. The BF600 always works as GAP peripheral / GATT server.

Unless specified otherwise, all fields are in the order of LSO to MSO. Where LSO = Least Significant Octet and MSO = Most Significant Octet.

Please note there are several sequences of commands that have to be made for initialization and some that are recommended from us. Please check the sequence diagrams before you start implementing the scale.

1.1 Services and Characteristics

1.1.1 Services

Specification Name	Assigned Number	UUID	Note
Generic Access	0x1800	00001800-0000-1000-8000-00805f9b34fb	
Generic Attribute	0x1801	00001802-0000-1000-8000-00805f9b34fb	
Device Information	0x180A	0000180a-0000-1000-8000-00805f9b34fb	
Custom Service	0xFEBA	0000feba-0000-1000-8000-00805f9b34fb	
Battery Service	0x180F	0000180f-0000-1000-8000-00805f9b34fb	
Custom Service	0xFFFF0	0000fff0-0000-1000-8000-00805f9b34fb	
Body Composition Service	0x181B	0000181B-0000-1000-8000-00805f9b34fb	
Weight Scale Service	0x181D	0000181D-0000-1000-8000-00805f9b34fb	
Current Time Service	0x1805	00001805-0000-1000-8000-00805f9b34fb	
User Data	0x181C	0000181c-0000-1000-8000-00805f9b34fb	

1.1.2 Service Characteristics

Generic Access

For further information please check the documentation on the official Bluetooth website.

Name	UUID	Properties	Note
Device Name	0x2A00	Read	"BF600"
Appearance	0x2A01	Read	"17"-Unknown
Reconnection Address	0x2A03	Write	
Peripheral Privacy Flag	0x2A02	Read	Disabled
Peripheral Preferred Connection Parameters	0x2A04	Read	Connection Interval: 500.00-1100.00 ms Slave Latency:0 Supervision Timeout Multiplier: 152

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Generic Attribute

For further information please check the documentation on the official Bluetooth website.

Name	UUID	Properties	Note
Service Changed	0x2A05	Read, Indicate	

Device Information

For further information please check the documentation on the official Bluetooth website.

Name	Assigned Number	Properties	Note
Serial Number String	0x2A25	Read	“”
Manufacturer Name String	0x2A29	Read	“Beurer GmbH”
System ID	0x2A23	Read	
Firmware Revision String	0x2A26	Read	“1.0”
Model Number String	0x2A24	Read	“BF600”
Hardware Revision String	0x2A27	Read	“1.0”
Software Revision String	0x2A28	Read	“1.0”
PnP ID	0x2A50	Read	Bluetooth SIG Company: AMICCON Electronics Corporation: 0x00C0 Product ID: 0 Product Version: 0
IEEE 11073-20601 Regularly Certification Data List	0x2A2A	Read	

Custom Service (0xFEBA)

This service was used for firmware updates during the development and is not needed for the communication with the scale.

Name	UUID	Properties	Note
Custom characteristic	0xFA10	Notify, Write no response	
Custom characteristic	0xFA11	Indicate, Write	
Custom characteristic	0xFA13	Notify	

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Battery Service

For further information please check the documentation on the official Bluetooth website.

Name	UUID	Properties	Note
Battery Level	0x2A19	Read, Notify	

Custom Service (0xFFFD)

This Service is used to get and set scale settings and information that are not covered by the Bluetooth SIG Weight Scale, Body Composition and User Data Services.

Name	UUID	Properties	Note
Custom characteristic	0xFFFD1	Read, Write	Scale settings
Custom characteristic	0xFFFD2	Notify, Write	User List
Custom characteristic	0xFFFD3	Read, Write	Activity level
Custom characteristic	0xFFFD4	Notify, Write	Take measurement
Custom characteristic	0xFFFD5	Read	Reference Weight

Body Composition Service

The Body Composition Service will send additional weight measurement information like body fat, muscle etc.. There will be a short info in this document. For further information please check the documentation on the official Bluetooth website.

Name	UUID	Properties	Note
Body Composition Feature	0x2A9B	Read	
Body Composition Measurement	0x2A9C	Indicate	

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Weight Scale Service

The Weight Scale Service will send the weight measurement information. There will be a short info in this document. For further information please check the documentation on the official Bluetooth website.

Name	UUID	Properties	Note
Weight Scale Feature	0x2A9E	Read	
Weight Measurement	0x2A9D	Indicate	

Current Time Service

The current time Service is used to set or read the time on the scale. There will be a short info in this document. For further information please check the documentation on the official Bluetooth website.

Name	UUID	Properties	Note
Current Time	0x2A2B	Read, Notify, Write	
Local Time Information	0x2A0F	Read	
Reference Time Information	0x2A14	Read	

User Data

The user data is used to manager the users on the scale. There will be a short info in this document. For further information please check the documentation on the official Bluetooth website.

Name	UUID	Properties	Note
Date of Birth	0x2A85	Read, Write	
Gender	0x2A8C	Read, Write	
Height	0x2A8E	Read, Write	
Database Change Increment	0x2A99	Read, Write, Notify	

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User Index	0x2A9A	Read	
User Control Point	0x2A9F	Write, Indicate	

1.2 Advertising Package

The advertising package of the BF600 only uses Bluetooth SIG standard advertising data and don't contain any company specific data (see table below). For further information please check the Bluetooth documentation.

Name	Data Type Value	Length	Description
Flags	0x01	2 byte	"0x06" General Discoverable BR/EDR is not supported
Incomplete List of 16-bit Service UUIDs	0x02	3 byte	"0x1D18" Weight Scale Service
Appearance	0x19	3 byte	"0x800C" Generic Weight Scale
LE BT Device Address	0x1B	7 byte	Contains the BLE MAC address of the device
Service Data	0x16	3-11 byte	The first two data byte are the weight scale Service "0x1D18" followed by a list with all users that have new measurements (not downloaded by any device). For example if user 1 and user 3 has new measurements the data would be "0x1D180103"
Complete Local Name	0x09	6 byte	"0x4246363030" BF600 in ASCII

1.3 Detailed Service Information

In this chapter some of the Services are explained. Only Services that are not Bluetooth confirm or very important are explained here. Please also check the Bluetooth documentation if you are not familiar with it.

1.3.1 Current Time Service

After turning on the BF600 the first time or after a battery change it shows "APP" flashing on the display. This means that scale doesn't have a system time and it is necessary to set the current time. To set the time you have to write it to the Current Time Characteristic (0x2A2B) with the following format:

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Current Time (Read Write Notify)

Name	Format	Info	Note
Year	2 byte	The year of the Gregorian calendar. Valid values are 1582-9999. 0 Means the year is unknown.	
Month	1 byte	The current Month. Valid values are 0-12. 0 = unknown 1 = January 2 = February ...	
Day	1 byte	The current Day. Valid values are 0-31. 0 = unknown 1 = First day of the Month 2 = Second day of the Month ...	
Hours	1 byte	The current hour of the day. Valid values are 0-23. 0 = 00:xx o'clock 1 = 01:xx o'clock ...	
Minutes	1 byte	The current minute. Valid values are 0-59. 0 = xx:00 o'clock 1 = xx:01 o'clock ...	
Seconds	1 byte	The current second. Valid values are 0-59. 0 = xx:xx:00 o'clock 1 = xx:xx:01 o'clock ...	
Day of the week	1 byte	The day of the week. This is not used by the scale, so you can set it to 0 (unknown day of the week).	
Fraction 256	1 byte	The 256 th fraction of a second. It is not used by the BF600, please set it to 0.	
Adjust Reason	1 byte	The adjust reason is not used by the BF600, so please set it to 0.	

After the time has been set the scale can be operated and it will create a timestamp for every measurement. It is suggested to check and set the time in regular intervals (for example on every connection), so the scale time doesn't differ too much from the smartphone time. It is also important that the smartphone has a valid time.

Local Time Information (Read)

Doesn't contain any useful information, please do not use.

Reference Time Information (Read)

Doesn't contain any useful information, please do not use.

1.3.2 User Data

The user data are used to read or write the data of a specific user. Before you can read or write any user data, you have to use the user index and consent code to login with the user. Otherwise it is not possible to read or write any data. There are some additional user data values in the Custom Service (0xFFFO) that are not supported by the Bluetooth SIG Standard. To read or write these values you also need to assign a user with the User Control Point via Consent Code. Here are the Characteristics in the User Data and a small description:

Date of Birth (Read Write)

Name	Format	Info	Note
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Year	2 byte	The year of birth in the Gregorian calendar. Valid values are 1582-9999. 0 Means the year is unknown. 0xFF FF means no user selected	
Month	1 byte	The Month of birth. Valid values are 0-12. 0 = unknown 1 = January 2 = February ... 0xFF means no user selected	
Day	1 byte	The Day of birth. Valid values are 0-31. 0 = unknown 1 = First day of the Month 2 = Second day of the Month ... 0xFF means no user selected	

Gender (Read Write)

Name	Format	Info	Note
Gender	1 byte	0 = male 1 = female 2 = unspecified 0xFF = no user selected	

Height (Read Write)

Name	Format	Info	Note
Height	1 byte	The height in meter with a resolution of 0,01 meter.	

Database Change Increment (Read Write Notify)

Name	Format	Info	Note
Increment	4 byte	Should be increased whenever some user data have been changed. Sends 0xFF FF FF FF if no user is selected. If the user data on the scale are updated during a connection, the scale will send a Notification. After 0xFF FF FF FE has been reached the next value should be 0x 00 00 00 00 again. IF the user data on the scale has been changed the scale will increment this value as well.	

User Index (Read)

Name	Format	Info	Note
User Index	1 byte	The index of the current User. Valid values are 1-8. 1= User 1 2 = User 2 ... Sends 0xFF if no user is selected/ unknown user.	

User Control Point (Write Indicate)

Name	Format	Info	Note
OP Code	1 byte	The OP Code is used to define the command that is send to the User Control Point	
Parameter	0-18 byte	The Parameters are used to transfer additional information for certain command to the User Control Point	

There are three commands defined by the Bluetooth SIG that are also used by the BF600. The first one is the “create new user” command:

Name	Format	Info	Note
OP Code	1 byte	The OP Code to create a new user is 0x01	
Parameter	2 byte	If a new user is created a consent code has to be send as a parameter.	

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		The consent code is a number between 0-9999 (0x0000 to 0x270F) and is needed to login this user.	
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After the “create new user” command has been sent, the BF600 will create a user index (if there is a free user slot) and save the consent code to this user. The user will have default values and they have to be changed after the user creation. To make measurements, read or change the user data you have to assign the user with the “consent procedure” after the user was created. There will be the following response from the BF600 via indication:

Name	Format	Info	Note
Response OP Code	1 byte	The OP Code for a response of the scale is always 0x20	
Request OP Code	1 byte	The OP Code of the create new user command, always 0x01	
Response Value	1 byte	0x01 = Success 0x02 = OP Code not supported 0x03 = Invalid Parameter 0x04 = Operation failed	
Response Parameter	1 byte	The User Index. This value is only present if the command was successful (Response value 0x01)	

The second command of the User Control Point is the “consent procedure” to assign a user:

Name	Format	Info	Note
OP Code	1 byte	The OP Code to create a new user is 0x02	
Parameter	3 byte	The first byte is the user index of the user that should be assigned. The other two byte are the consent code of the user (in order LSO to MSO).	

If the user has been assigned successfully, it is possible to read the User data and some user specific data of the Custom Service (0xFF0). There will be a response via indication if the command has succeeded or an error has occurred:

Name	Format	Info	Note
Response OP Code	1 byte	The OP Code for a response of the scale is always 0x20	
Request OP Code	1 byte	The OP Code of the consent procedure command , always 0x02	
Response Value	1 byte	0x01 = Success 0x02 = OP Code not supported 0x03 = Invalid Parameter 0x04 = Operation failed 0x05 = User not Authorized	

The third user control point command is the “delete user data” command:

Name	Format	Info	Note
OP Code	1 byte	The OP Code to delete a user is 0x03. This command will delete the user that is currently assigned to the scale with the consent procedure.	

The BF600 will delete the data of the current user as well as the consent code. The user index characteristic is set to 0xFF (no user selected/unknown user). The scale will send the following indication after the command has processed:

Name	Format	Info	Note
Response OP Code	1 byte	The OP Code for a response of the scale is always 0x20	
Request OP Code	1 byte	The OP Code of the delete user data command , always 0x03	

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Response Value	1 byte	0x01 = Success 0x02 = OP Code not supported 0x04 = Operation failed 0x05 = User not Authorized (will also occur if no user was selected when the "delete user data" command was sent)	
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1.3.3 Weight Scale Service

The Weight Scale Service provides the basic weight measurement data of a user. Data like Body Fat, BMR, Body Water Mass etc. are not part of the Weight Scale Service and can be found in the Body Composition Service. The Weight Measurements are sent as soon as a user is assigned via consent code and new measurements are present.

The Weight Scale Service is implemented according to the Bluetooth SIG Standard. There will be a short explanation of the characteristics below. If you need further information please check the official Bluetooth SIG documentation.

Weight Scale Feature (Read)

Name	Format	Info	Note
Weight Scale Feature	4 byte	The value is 0xB7 00 00 00. That means the measurements support a timestamp, multiple users, BMI and has a resolution of 0.01kg for weight and 0.01 m for length.	

Weight Measurement (Indicate)

Name	Format	Info	Note
Flags	1 byte	0x0E	
Weight	2 byte	The weight with a resolution of 0.01 kg.	
Year	2 byte	The year of the measurement	
Month	1 byte	The month of the measurement	
Day	1 byte	The day of the measurement	
Hour	1 byte	The hour of the measurement	
Minute	1 byte	The minute of the measurement	
Second	1 byte	The second of the measurement	
User ID	1 byte	The user ID of the measurement	
BMI	2 byte	The BMI of the user at the time of the measurement	
Height	2 byte	The height of the user at the time of the measurement	

1.3.4 Body Composition Service

The Body Composition Service provides additional information to the weight data of the weight scale service. The indication of the Body Composition measurement is directly sent after the according weight measurement. If there is a measurement with only weight measurements the Body Composition Measurement is also sent by the device, but all the values that are not present are set to 0.

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Body Composition Feature (Read)

Name	Format	Info	Note
Body Composition Feature	4 byte	The value is 0xCF 31 00 00. That means the measurements support a timestamp, multiple users, BMR, Muscle in %, Soft Lean Mass, Body Water Mass, Impedance and a resolution of 0.01 kg	

Body Composition Measurement (Indicate)

Name	Format	Info	Note
Flags	1 byte	0x98 03	
Body Fat %	2 byte	The Body Fat value in %.	
BMR	2 byte	The BMR value in kcal	
Muscle %	2 byte	The Muscle value in %	
Soft Lean Mass	2 byte	The Soft Lean Mass in 0.01 kg	
Body Water Mass	2 byte	The Body Water Mass in 0.01 kg	
Impedance	2 byte	The impedance in Ohm	

1.3.5 Custom Service (0xFF0)

In the Custom Service are features that are not part of the Bluetooth SIG documentation. Some of the characteristics contain information to the assigned user, additional measurement information or additional commands.

User List 0xFF2 (Write/Notify)

Use this command to ask for a List of the users on the BF600. It is not necessary to assign a user via consent code to get the User List. Send the following write command to the Scale to get the User List:

Name	Format	Info	Note
Get User List	1 byte	0x00 has to be sent to ask for the User List	

The scale will send two kinds of notifications. The first one contains the data of a user and will be sent for every user that is present on the scale:

Name	Format	Info	Note
Op Code	1 byte	0x00 for user information	
User Index	1 byte	The Index of the User	
Initials	3 byte	The Initials of the User. The BF600 doesn't use this feature and it is shown 0xFF FF FF.	
Year of Birth	2 byte	The year when the user was born	
Month of Birth	1 byte	The month when the user was born	
Day of Birth	1 byte	The day when the user was born	
Height	1 byte	The height of the user	
Gender	1 byte	The gender of the user	

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Activity Level	1 byte	The Activity Level of the user	
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The BF600 will send another notification with the status of the command:

Name	Format	Info	Note
Command Status	1 byte	0x01 Success and User List data transfer complete 0x02 No user present on the scale	

Scale Setting 0xFF1 (Read/Write)

The Scale Settings are used to read the current settings on the scale or to change them with a write command. They are general scale settings and apply for all users.

Name	Format	Info	Note
Language	1 byte	The selected language. This option is not supported on the BF600. The value should be 0xFF and not be changed.	
Unit	1 byte	The unit of the scale. 0x00 = kg 0x01 lb 0x02 st.	
Time Format	1 byte	The time format. This option is not supported on the BF600. The value should be 0xFF and not be changed.	
Guest Mode	1 byte	Activates or deactivates the guest measurement mode. This option is not supported on the BF600. The value should be 0xFF and not be changed.	
Weight Threshold	2 byte	The weight threshold to assign a measurement to the according user. The resolution is in 0.1 kg and the valid range is from 0.1 kg-25.5 kg. The default value is ± 3.0 kg.	
Body Fat Threshold	2 byte	The body fat threshold to assign a measurement to the according user. This value is not used for the user assignment on the BF600.	

Take Measurement 0xFF4 (Write/Notify)

Use this command to trigger a measurement on the current user. This command only works if a user is assigned via consent code. The Weight Threshold will be ignored for this measurement and it will be always assigned to the user. Use this write command to start the measurement:

Name	Format	Info	Note
Start Measurement	1 byte	0x00	

The scale will send a notification with the measurement status:

Name	Format	Info	Note
Measurement Status	1 byte	0x01 = Success 0x02 = Take Measurement Timeout 0x03 = Take Measurement Failed 0x04 = No user selected	

Activity Level 0xFF3 (Read/Write)

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Use this command to write or read the activity data of the selected user. This command will only work if a user is selected.

Name	Format	Info	Note
Measurement Status	1 byte	0x01 = Activity Level 1 0x02 = Activity Level 2 0x03 = Activity Level 3 (default value) 0x04 = Activity Level 4 0x05 = Activity Level 5 0xFF= No user selected (only Read command)	

Reference Weight 0x FFF5 (Read)

Read the Reference Weight of the selected user. This command will not work if no user is selected.

Name	Format	Info	Note
Reference Weight	2 byte	The Reference weight with a resolution of 0.01 kg. If no user is selected it will show 0xFF FF.	
Reference Body Fat	2 byte	This value is not used in the BF600 and always 0xFF FF	

1.4 Sequence diagrams

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